A proof of the conjectured linear recurrence for OEIS A184491

Adrian Perez Fontelles and Gaspard Moulinier, Independent researchers

09 September 2026

1 The conjecture

The Online Encyclopedia of Integer Sequences records A184491 as

> *Number of (n+1)X4 0..2 arrays with every 2X2 subblock sum greater than 4*

and carries in its Formula section the line:

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Empirical:  a(n)=14*a(n-1)+235*a(n-2)-835*a(n-3)-9872*a(n-4)+18170*a(n-5)+157753*a(n-6)-
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That line carries no separate signature; the entry itself is by R. H. Hardin (Jan 15 2011).

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 9, last modified Oct 03 2025, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 14\,a(n-1)+235\,a(n-2)-835\,a(n-3)-9872\,a(n-4)+18170\,a(n-5)+157753\,a(n-6)-198571\,a(n-7)-1173439\,a(n-8)+1189558\,a(n-9)+4436018\,a(n-10)-3898484\,a(n-11)-9184174\,a(n-12)+7095216\,a(n-13)+11019526\,a(n-14)-7342318\,a(n-15)-7892736\,a(n-16)+4325550\,a(n-17)+3382616\,a(n-18)-1396572\,a(n-19)-837900\,a(n-20)+220320\,a(n-21)+107712\,a(n-22)-12096\,a(n-23)-5184\,a(n-24).$$

stated without a range; it first has meaning at $n = 25$, the least index at which all of $a(n-1), \ldots, a(n-24)$ are terms of the sequence.

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 4$ array over $\{0, \ldots, 2\}$ with $n' = n + 1$ rows. For $0 \le i < n' - 1$ and $0 \le j < 3$ write $B_{i,j}$ for the 2×2 subblock with entries $a = A[i][j]$, $b = A[i][j{+}1]$, $c = A[i{+}1][j]$, $d = A[i{+}1][j{+}1]$.

The statistic the entry names, *sum*, is written $s(B)$ below.

Every subblock has $s(B)$ greater than one of $\{4\}$.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

A 2×2 subblock lies in two consecutive rows, and the conditions that compare a subblock with a neighbour reach one row further, so everything is decided by three consecutive rows. Take as state the last two rows; appending a row decides every condition whose lowest row is the new one.

An array of n' rows is then exactly a walk of length n' from the empty state, and every array arises from exactly one walk, so $a(n) = w^{\mathsf{T}} M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 874 states, 874 after trimming; the forward identification of states with equal numbers of completions of every length, then the mirror identification on the edges coming in, leaves $S = 26$ blocks, and the count satisfies the linear recurrence given by the characteristic polynomial of that 26×26 matrix. The bound is computed, not assumed.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{\,m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 26: writing that polynomial as $t^{26} - \sum_{i=1}^{26} \lambda_i t^{26-i}$ gives $u_{m+26} = \sum_{i=1}^{26} \lambda_i u_{m+26-i}$. Hence 26 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A184491, the recurrence of order* 24 *displayed in Section 1 holds for every* $n \geq 25$*, at every index at which it can be stated.*

Proof. The residuals u_m were computed exactly for $m = 1, \ldots, 70$. Every one of them vanished. After it, more than $S = 26$ consecutive residuals vanish, so by Section 5 every later residual vanishes as well. $\square$

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 14 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 792, 20514, 421879, 9573698, 209361859, 4649028105, 102612963762, 2270452765634, ...
- **An independent enumeration.** For $n = 1, \ldots, 2$ every one of the 3^{4n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 26$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A184491, revision 9, last modified Oct 03 2025.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.
4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.